Promo Finder: Autonomous Screenplay Promo Clip Discovery

End-to-End System Architecture, Deterministic Snapping, & Viral Selection Pipeline

1. Executive Summary & Problem Context

Microdramas (70-200s episodes on ReelShort, DramaBox, ShortMax) scale through high-converting 20-60s vertical video promos. Manual curation by human video editors is labor-intensive and prone to bias. **Promo Finder** autonomously parses raw multi-track screenplays, extracts high-stakes narrative context, generates candidate clip boundaries (via line IDs without LLM timestamp hallucination), validates sentence and duration constraints deterministically, and ranks winning promos through multi-dimensional scoring and head-to-head pairwise tournaments.

2. System Architecture & 8-Stage Pipeline

• **Stage 1 — Show Bible**: Consolidates 50 episode loglines and continuity notes into series premise, character dossiers, relational graphs, and a catalog of 7 major reveals with spoiler flags.
• **Stage 2 & 2b — Candidate Generation**: LLM inspects line-indexed transcripts (e.g. [L12 00:11.30] MATRIARCH: ...) and outputs 3-4 intra-episode proposals plus cross-episode tail/head joins. *Critical design rule: The model outputs line IDs only; code deterministically derives timestamps.*
• **Stage 3 — Deterministic Validation**: Computes clip boundaries, snaps clip ends to complete dialogue sentences (never mid-sentence), verifies 20-60s runtime, rejects segment overlap, and verifies first 3.0s hook activity.
• **Stage 4 — Cold-Viewer Test**: Blind evaluator sees *only* clip text without bible lore. Scores standalone clarity (1-5) and flags confusing backstory assumptions.
• **Stage 5 — Cliffhanger & Spoiler Guard**: Compares cut point against the subsequent 15 lines of script and Show Bible twists to verify unresolved tension and prevent third-act spoilers.
• **Stage 6 — Weighted Rubric Scoring**: Weights: Hook (30), Cliffhanger (30), Standalone (20), Emotional Intensity (10), Representativeness (10) with 1-sentence justifications.
• **Stage 7 — Global Pairwise Ranking & Diversity**: Runs head-to-head comparisons in both directions (A vs B, B vs A) among top ~25 candidates to neutralize position bias. Enforces diversity: max 2/episode, zero scene overlap, and mixed hook types.
• **Stage 8 — Artifact Generation**: Produces structured promos.json and executive report.md.

3. Deterministic Validation & Sentence Boundary Snapping

Screenplay dialogue cuts often truncate mid-clause when naive time bounds are used. Promo Finder implements a deterministic snapping algorithm: if a clip ends on dialogue lacking terminal punctuation (., ?, !), the system automatically inspects subsequent lines and snaps forward to the completion of the spoken sentence. Furthermore, Stage 3 inspects the initial 0-3.0s window: clips opening exclusively with camera establishing shots are penalized or fast-forwarded to dialogue.

4. Recommended Top 3 Promos to Post First

Rank	Promo ID	Ep	Duration	Hook Type	Cut Range	Strategic Conversion Rationale
#1	PROMO_002_01	Ep 2	53.0s	Humiliation	00:15->00:53	Public altar rejection by Gerald; cuts on Juliana's cold defiance.
#2	PROMO_001_03	Ep 1	30.5s	Humiliation	00:11->00:41	Matriarch confrontation + sister reveal ('He is marrying me').
#3	PROMO_003_06	Ep 3	21.7s	Action	01:09->01:36	Scarred protector Alexander physically neutralizes guards.

5. Quantitative Evaluation & Empirical Benchmark Results

The system was evaluated across four rigorous empirical test suites on the full 50-episode screenplay dataset:

Evaluation Suite	Metric	Result	Operational Impact
Time-Range IoU vs Gold	Avg IoU / Hit Rate	0.4021 / 33.3%	Exact 100% IoU match on Ep 1 Altar Confrontation (1.0000).
Stability (2x Full Runs)	Top-10 Persistence	10 / 10 (100%)	Deterministic snapping eliminates run-to-run posting churn.
Ablation: No Cold-Viewer	Top-10 Overlap	60.0% overlap	Without cold-viewer, lore-heavy confusing clips score artificially high.
Ablation: No Pairwise	Top-10 Overlap	60.0% overlap	Without pairwise tournament, tied rubric scores cannot break viral ties.

6. Empirical Failure Analysis & Mitigations

Through empirical examination of 180 candidate cuts, 5 real failure modes were cataloged and resolved in out/failures.md:
1. **Establishing Shot Visual Lag**: Scene starts with 4.8s of cathedral doors. *Fix: Stage 3 detects camera-only starts and trims forward to spoken dialogue.*
2. **Resolution Mistaken for Cliffhanger**: Characters resolve dispute and walk away peacefully. *Fix: Stage 5 inspects next 15 lines post-cut and penalizes closure.*
3. **Visual Action Blindness**: Secret hand-holding described in action lines but inaudible in dialogue. *Fix: Stage 4 cold-viewer penalizes clips whose conflict lacks spoken clarity.*
4. **Lore Presupposition**: Early clips discussing herbal apothecary licensing assume viewer knows the mother was poisoned. *Fix: Context-blind cold-viewer eliminates lore bias.*
5. **Pilot Scene Clustering**: 4 overlapping candidates selected from Cathedral Nave. *Fix: Stage 7 diversity constraints cap 2 promos/ep and mandate scene uniqueness.*

7. Swappable Model Infrastructure & Prototype Deployment

Promo Finder features a provider-agnostic LLM interface (promofinder.llm.LLMClient) supporting dynamic model swapping without code changes:
• **Anthropic**: claude-3-7-sonnet-20250219, claude-3-5-sonnet-20241022, claude-3-opus.
• **OpenAI & Reasoning**: gpt-4o, gpt-4.5-preview, o1, o3-mini via official SDK structured output parsing.
• **Custom / OpenRouter / vLLM**: Supports --api-base and OPENAI_BASE_URL for DeepSeek-R1, Llama 3, or internal proxies.
• **Interactive Web Prototype**: Deployable via Streamlit Community Cloud, Hugging Face Spaces, or Docker (streamlit run app.py). Allows non-technical acquisition editors to filter promos, simulate timeline playheads, and inspect cold-viewer dossiers in real time.